Asphyxia

Forensic Medicine | 3rd Year MBBS

Hi Sammy!
Asphyxia = one of the
HEAVIEST FMT topics!
Focus on comparisons!

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Definition, Types + General Features

Asphyxia = oxygen lack leading to death. Know all types and signs!

ASPHYXIA

"Condition caused by interference with respiration or lack of oxygen in inspired air." — Reddy

Classification of Asphyxia

Category	Types
Hanging	Suicidal (most common), homicidal, accidental, judicial
Strangulation	Ligature strangulation, throttling (manual), bansdola, garrotting
Suffocation	Smothering (mouth/nose closed), choking (foreign body), overlaying
Drowning	Wet drowning, dry drowning, near-drowning, immersion syndrome
Traumatic asphyxia	Crush asphyxia (stampede, burial, building collapse)
Environmental	Confined space (CO_2 excess, O_2 deficiency), vitiated atmosphere

General Post-mortem Features of Asphyxia

EXTERNAL

- Cyanosis (bluish discoloration)
- Congestion of face & neck
- Petechial hemorrhages (Tardieu spots)
- Edema of face
- Protruding tongue, eyes

INTERNAL (Autopsy)

- Visceral congestion (lungs, brain)
- Tardieu spots on lungs, heart
- Dark fluid blood (deoxygenated)
- Right heart distended
- Pulmonary edema

Tardieu Spots!

Petechial hemorrhages (1-2mm) on pleura, pericardium, thymus due to rupture of capillaries from increased venous pressure. Classic sign of asphyxia!

This comparison table is THE most important in all of asphyxia!

Hanging vs Ligature Strangulation

Feature	Hanging	Ligature Strangulation
Force applied by	Body weight (gravity)	Other person (manual)
Ligature mark	Oblique, above thyroid	Horizontal, at/below thyroid
Mark continuity	Incomplete (non-encircling)	Complete (encircling)
Knot position	Highest point (neck)	Any position
Face color	Pale (typical) / Congested	Congested, cyanosed
Petechiae	May be absent (typical)	Prominent (always present)
Tongue	Protruded, may be bitten	Protruded, swollen, cyanosed
Neck structures	Fractured hyoid (rare)	Fractured hyoid (common)
Cause of death	Asphyxia / vascular / spinal	Asphyxia
Common manner	Suicidal (most common)	Homicidal (most common)

Types of Hanging

Type	Description
Complete	Body fully suspended, feet off ground
Partial/Incomplete	Part of body touching ground (most common in suicide)
Typical	Knot at occiput (back of head) — face pale
Atypical	Knot at any other position — face congested
Judicial	Long drop method — cervical spine fracture (C2 hangman's)

Throttling (Manual Strangulation)

- Compression of neck by hand/fingers — ALWAYS homicidal
- Nail marks (crescentic abrasions) on neck — pathognomonic
- Hyoid bone fracture more common than in hanging
- Finger-tip bruises on neck (both sides) — classic finding

Drowning — Types + Autopsy Findings

Fresh vs salt water
drowning + diatom test
= super high yield!

Types of Drowning

Type	Description
Wet drowning	Water enters lungs — most common type (90%)
Dry drowning	Laryngospasm prevents water entry — 10% cases
Secondary drowning	Death hours/days after rescue due to pulmonary edema
Immersion syndrome	Sudden death on contact with cold water — vagal reflex
Near-drowning	Survival >24 hours after submersion episode

Fresh Water vs Salt Water Drowning

Feature	Fresh Water	Salt Water
Osmolarity	Hypotonic to blood	Hypertonic to blood
Fluid movement	Water enters blood	Blood enters alveoli
Blood volume	Hemodilution (increased)	Hemoconcentration (decreased)
Electrolytes	Diluted (hyponatremia)	Concentrated (hypernatremia)
Cause of death	Ventricular fibrillation	Pulmonary edema, heart failure
Lungs	Voluminous, emphysematous	Heavy, waterlogged
Death onset	Faster (4-5 min)	Slower (8-12 min)

Autopsy Findings in Drowning

Finding	Detail
Fine white froth	At mouth & nostrils (mushroom-shaped) — classic sign
Emphysema aquosum	Voluminous, edematous lungs with rib markings
Water in stomach	Swallowed water (>500ml suggests ante-mortem)
Cutis anserina	Goose skin (post-mortem, not specific)
Washerwoman hands	Wrinkling of palms/soles (prolonged immersion)

Diatom Test!
Diatoms = unicellular algae with silica shell. Found in bone marrow/organs = proof of ante-mortem drowning (entered via lungs -> blood -> organs). Not 100% reliable alone.

Exam Strategy + High Yield

Asphyxia = guaranteed
LAQ every single year!
You've GOT this Sammy!

Most Repeated LAQs (10–12 marks)

1. Differentiate hanging from ligature strangulation (table format)
2. Describe types of drowning. Give autopsy findings in fresh water drowning
3. Define asphyxia. Classify and describe general features of asphyxial death
4. Discuss post-mortem findings in hanging with medicolegal importance
5. Compare fresh water vs salt water drowning with mechanism of death

Most Repeated SAQs (5–6 marks)

1. Diatom test — principle, method, significance
2. Throttling — definition and findings
3. Tardieu spots — definition and significance
4. Types of hanging (complete/partial, typical/atypical)
5. Dry drowning and immersion syndrome
6. Cafe coronary (choking death during eating)

One-Liner Associations (MCQ/Viva Gold)

Question	Answer
Most common manner of hanging	Suicidal
Most common manner of strangulation	Homicidal
Ligature mark in hanging	Oblique, above thyroid cartilage
Ligature mark in strangulation	Horizontal, at/below thyroid
Crescentic nail marks on neck	Throttling (manual strangulation)
Classic sign of drowning	Fine white froth at mouth/nose
Diatom test confirms	Ante-mortem drowning
Hyoid fracture — more common in	Strangulation (>40yr age)
Judicial hanging fracture	C2 (hangman's fracture)
Most common type of drowning	Wet drowning (90%)

♥ **Understanding these deaths helps prevent them, Sammy!**
Your knowledge will help uncover truth and deliver justice!